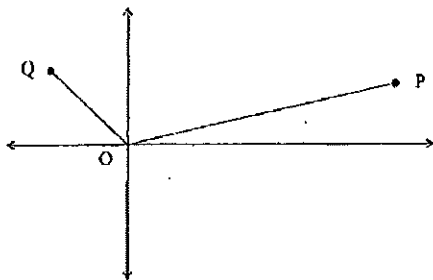


Question 2

Marks

- (a) The diagram shows a complex plane with origin O.



Points P and Q represent non-zero complex numbers z and w respectively.

- | | |
|---|---|
| (i) Write down the length of PQ in terms of z and w . | 1 |
| (ii) Copy the diagram into your booklet. Construct point R that represents $z + w$.
What type of quadrilateral is OPRQ? | 2 |
| (iii) Prove that if $ z + w = z - w $, the complex number $\frac{w}{z}$ is imaginary. | 2 |
- (b) Given ω is a complex root of $z^3 - 1 = 0$;
- | | |
|--|---|
| (i) Explain carefully why ω^2 is also a complex root of $z^3 - 1 = 0$. | 2 |
| (ii) Prove that $1 + \omega + \omega^2 = 0$ | 1 |
| (iii) Show the roots of $z^3 - 1 = 0$ on an Argand diagram. | 1 |
| (iv) Simplify $(1 + \omega^2)^{2012}$ completely. | 1 |
- (c) Suppose that $H(n) = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$, for $n = 1, 2, 3, \dots$
- So $H(1) = 1$, $H(2) = 1 + \frac{1}{2}$, $H(3) = 1 + \frac{1}{2} + \frac{1}{3}$, and so on.
- Prove by mathematical induction that
- $$n + H(1) + H(2) + H(3) + \cdots + H(n-1) = nH(n)$$
- for $n = 2, 3, 4, \dots$

End of Assessment



Year 12 Mathematics Extension 2
HSC ASSESSMENT TASK 1
Term 4 Week 8 2011

Name: _____

Teacher: _____

Wednesday 30th November 2011

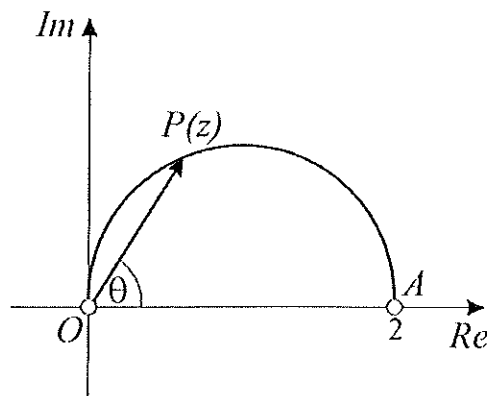
Set by: VUL

- Attempt **both** questions.
- Both questions are of equal value.
- Marks may be deducted for insufficient, or illegible work.
- Only Board approved calculators (**excluding** graphic calculators) may be used
- Total possible mark is 30
- **Begin each question on a new page.**
- **TIME ALLOWED : 45 minutes + 2 min reading time**

Question 1

Marks

- (a) Express $\frac{23 - 14i}{3 - 4i}$ in the form $a + bi$, where a and b are real. 2
- (b) Find the two square roots of $-16 + 30i$. 2
- (c) Let $w = -\sqrt{3} + i$.
- (i) Express w in modulus-argument form. 2
- (ii) Show that $w^9 + 512i = 0$. 2
- (d) Shade the region in the complex plane where $|z + 2| \leq 2$ and $-\frac{\pi}{6} \leq \arg(z + 3) \leq \frac{\pi}{3}$ are simultaneously satisfied. 3
- (e)



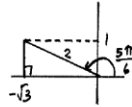
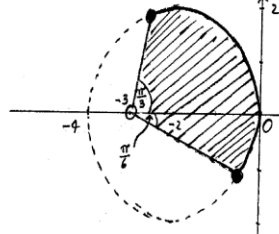
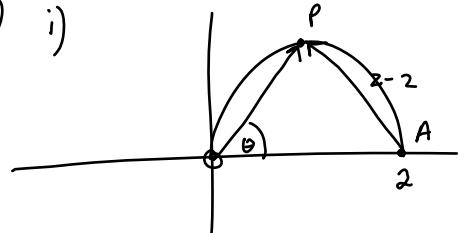
The diagram above shows the semicircular locus of the point P that represents the complex number z .

Let $\arg z = \theta$, as shown on the diagram.

- (i) Copy the diagram and on it show a vector representing $z - 2$. 1
- (ii) Explain why $\left| \frac{z - 2}{z} \right| = \tan \theta$. 1
- (iii) Show that $\arg \left(\frac{z - 2}{z} \right) = \frac{\pi}{2}$. 2

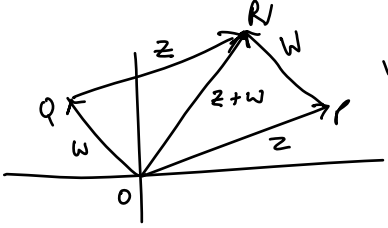
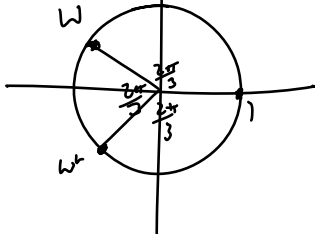


2012 Year 12 Mathematics Extension 2 HSC Task 1 SOLUTIONS

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
(a) $\frac{23-14i}{3-4i} \cdot \frac{3+4i}{3+4i} = \frac{69+56-42i+92i}{9-16i^2}$ ⁽³⁾ $= \frac{125+50i}{25}$ $= 5+2i$ ✓ (b) Let $-16+30i = (a+ib)^2$. $\therefore a^2-b^2 = -16$ and $ab = 15$ ✓ By inspection, $(a,b) = (3,5)$ or $(-3,-5)$. ✓ So the two square roots are $3+5i$ and $-3-5i$. (c) (i) $w = -\sqrt{3} + i$ $= 2\text{cis}\frac{5\pi}{6}$ ✓  (ii) $w^9 = (2\text{cis}\frac{5\pi}{6})^9$ $= 2^9\text{cis}\frac{15\pi}{2}$ $= 512\text{cis}\frac{3\pi}{2}$ ✓ $= 512(0-i)$ $= -512i$, so $w^9 + 512i = 0$. So w is a root of the equation $z^9 + 512i = 0$. (d)  ✓ Boundaries ✓ Shading ✓ Open circle		e) i)  ii) Since $\angle OPA = 90^\circ$ ($\angle$ in a semi circle) $\tan \theta = \frac{ AP }{ OP }$ $= \frac{ z-2 }{ z }$ $= \left \frac{z-2}{z} \right$ iii) $\arg\left(\frac{z-2}{z}\right)$ $= \arg(z-2) - \arg(z)$ $= \angle PAx - \angle POA$ $= (\angle POA + \angle OPA) - \angle POA$ $= \angle OPA$ $= \frac{\pi}{2}$ x is positive x-axis (Ext $\angle$ of Δ equals sum of opp. $\angle$s)	



2012 Year 12 Mathematics Extension 2 HSC Task 1 SOLUTIONS

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
<u>Question 2</u> a) i) $\vec{PQ} = \vec{OQ} - \vec{OP}$ $= w - z$ ✓ ii)  ✓ $OPQR$ is a parallelogram ✓ iii) $z+w$ and $z-w$ represent the diagonals of quadrilateral $OPRQ$. If $z+w = z-w$ these diagonals are equal in length and $OPRQ$ is a rectangle ✓ $\therefore \vec{OQ} = ki \times \vec{OP}$ (rotation of 90° in an anticlockwise direction) ✓ so $w = ki z$ and $\frac{w}{z} = ki$ which is purely imaginary. c) When $n=2$ $LHS = 2 + H(1)$ $RHS = 2H(2)$ $= 2 + 1$ $= 2 \times (1 + \frac{1}{2})$ ✓ $= 3$ $= 2 \times \frac{3}{2} = 3$ ✓ $\therefore$ Statement true when $n=2$ Assume statement true for $n=k$ ie, $k + H(1) + H(2) + H(3) + \dots + H(k-1) = kH(k)$ When $n=k+1$ $L.H.S = k+1 + H(1) + H(2) + \dots + H(k-1) + H(k)$ $= 1 + kH(k) + H(k)$ by assumption ✓ $R.H.S = (k+1)H(k+1)$ $= (k+1) \times (1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{k} + \frac{1}{k+1})$ ✓ $= (k+1) \times [H(k) + \frac{1}{k+1}]$ ✓ $= (k+1)H(k) + 1$ $= kH(k) + H(k) + 1$		b) i) The complex cube roots of unity (ie, roots of $z^3=1$) are equally spaced $\frac{2\pi}{3}$ radians apart on the unit circle starting with $z=1$. So if roots $z_0=1$ then $z_1 = \text{cis } \frac{2\pi}{3} = w$ ✓ $z_2 = \text{cis } \frac{4\pi}{3} = (\text{cis } \frac{2\pi}{3})^2 = w^2$ or alternatively if $z^3=1$ $z^3-1=0$ $(z-1)(z^2+z+1)=0$ $\therefore z=1$ or $z = \frac{-1 \pm \sqrt{-3}}{2}$ $= \frac{-1 \pm \frac{\sqrt{3}}{2}i}{2}$ if $w = -\frac{1}{2} + \frac{\sqrt{3}}{2}i = \text{cis } \frac{2\pi}{3}$ then $-\frac{1}{2} - \frac{\sqrt{3}}{2}i = \text{cis } \frac{4\pi}{3}$ $= (\text{cis } \frac{2\pi}{3})^2 = w^2$ ii) $1+w+w^2 = \sum \text{of roots} = \frac{-b}{a} = 0$ ✓ iii)  ✓ iv) $(1+w^2)^{2012} = (-w)^{2012}$ $= w^{2012}$ $= (w^3)^{670} \times w^2$ $= 1^{670} \times w^2$ as $w^3=1$ ✓ $= w^2$	

$\therefore$ If true for $n=k$,
 true for $n=k+1$
 Since true for $n=2$
 true for $n=3, 4, 5, \dots$
 $\therefore$ True for all $n \geq 2 \in \mathbb{Z}$